

knowledge or consent. Even with a direct physical interface to the user, it is impractical for the cryptographic token to require user confirmation for every cryptographic operation.

This second vulnerability may be addressed through mandatory security, trusted path and protected path features in the operating system. A trusted path mechanism obviates the need for a separate physical interface for activation. A protected path mechanism permits the cryptographic token to identify its callers and enforce fine-grained controls over the use of services, algorithms, sessions and keys. As an alternative to having the token deal with fine-grained controls over its usage, mandatory security mechanisms may also be used to provide such controls. For example, mandatory security mechanisms may be used to isolate the token for use only by applications executed by the user who activated the token. Furthermore, the mandatory security mechanisms can reduce the risk of malicious software being able to use the cryptographic token and may consequently limit the use of the trusted path mechanism to highly sensitive actions.

Hence, even in the simplest case, the features of a secure operating system are crucial to addressing the vulnerabilities of application-space cryptography. In the remainder of this section, the assumptions of the simplified case are removed, and the additional vulnerabilities are examined.

If the assumption that initial keys are securely established within the token is removed, then there is the additional vulnerability that the initial keys may be observed or modified by an unauthorised entity. Unless the initial keys are provided via a dedicated physical interface to the cryptographic token, the operating system must protect the path between the initial key source and the cryptographic token and may need to protect the initial key source itself. Mandatory security mechanisms may be used to rigorously protect the path and the key source. A trusted path may be required for initial keying.

If the assumption that the cryptographic mechanism is con-